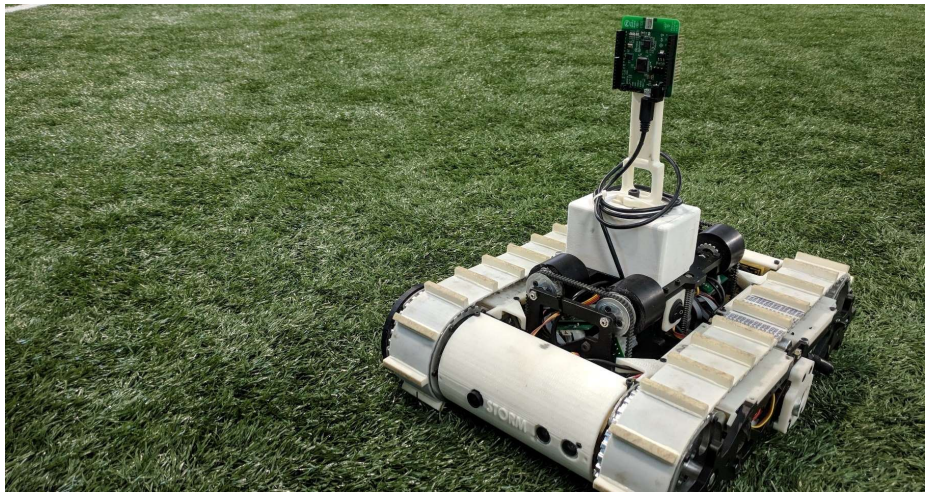


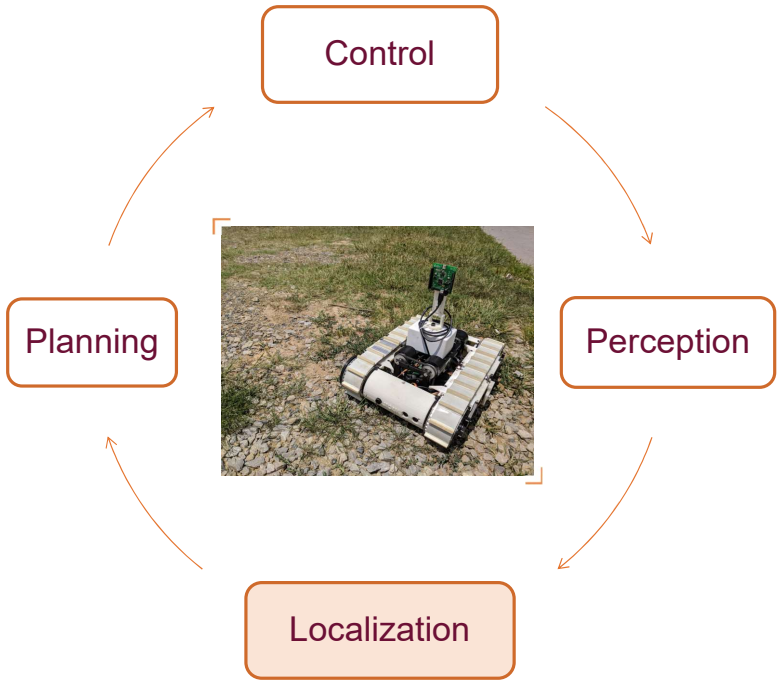
ED 5315
INTRODUCTION TO FIELD AND SERVICE ROBOTICS

Instructor: Bijo Sebastian

LOCALIZATION

Capabilities needed for field robotic systems:

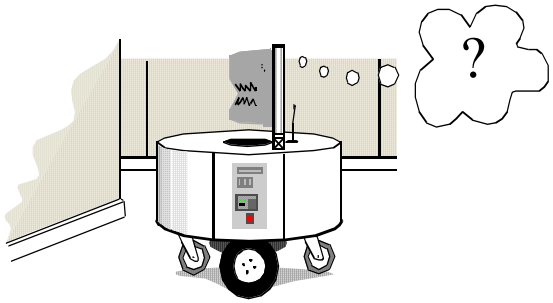
- Control: Locomotion - Mechanism, Kinematics, Dynamics, Actuator characteristics
- Perception: Sensor characteristics, uncertainty model, fusion
- **Localization: Algorithms, Probability theory**
- Planning: Artificial Intelligence



LOCALIZATION

The three key questions in Field/service Robotics

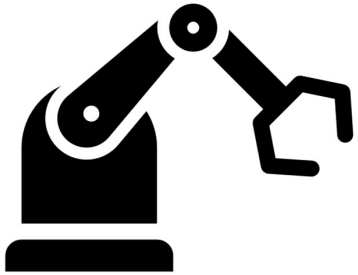
- Where am I ?
- Where am I going ?
- How do I get there ?

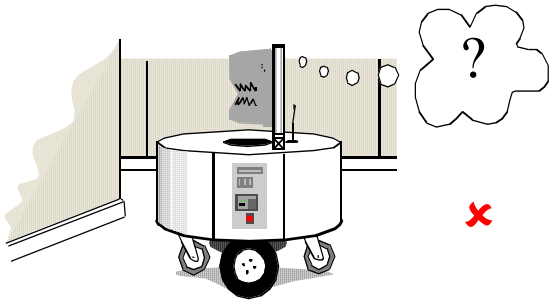


Requirements:

- Model of the environment (given or autonomously built)
- Capability to localize within the environment
- Capability to plan and execute motion within the environment

LOCALIZATION

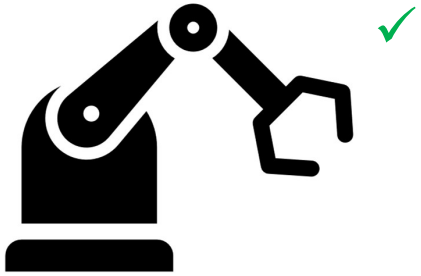


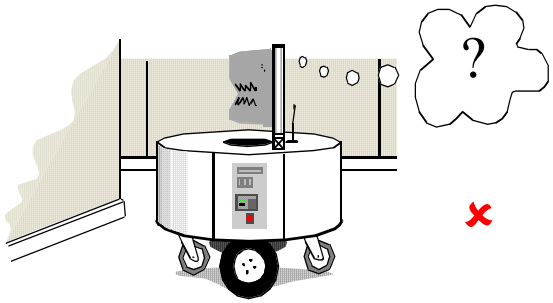


Localization is comparatively simpler for manipulators as compared to mobile robot

- Fixed base, known in global/inertial frame
- Encoders in each joint with minimal/no slip

LOCALIZATION





The following module will try to solve localization of a differential drive robot through sensor fusion:

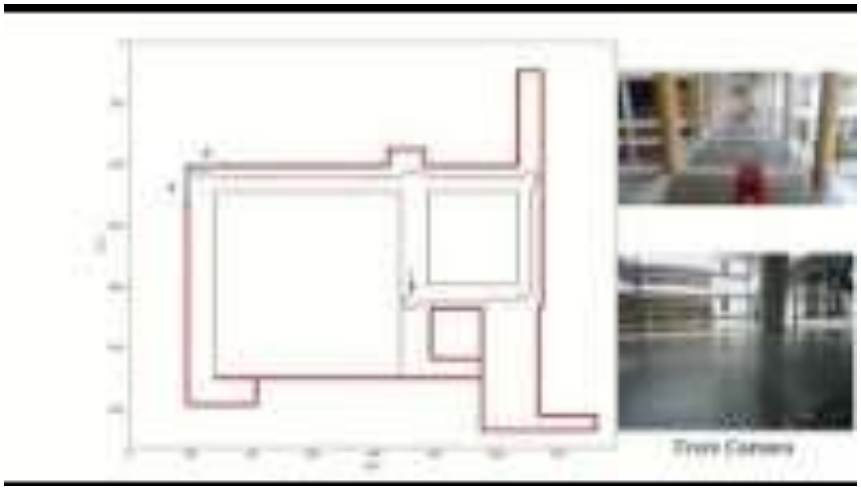
In doing so we will address the following questions:

- How to get an estimate when different sensors are reporting different values for the same measurement?
- Can we compare predicted sensor readings with actual sensor readings to improve robot pose estimation?

LOCALIZATION

Results from the implementation of Localization within Autonomous Systems Lab

The Implementation relies on Odometry, IMU, Lidar and Camera data , fusing all of them using an EKF



https://www.youtube.com/watch?v=69TlcWPwAac

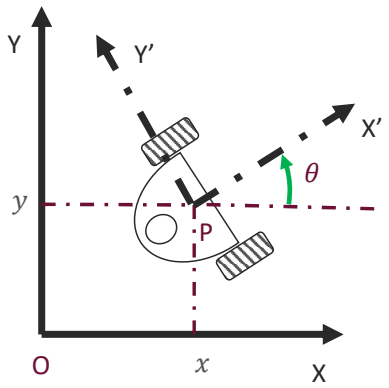
LOCALIZATION

States of mobile robot:

3 DOF system: x, y position and θ orientation

Pose of the robot (in global frame),

$$P = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}$$



Localization could be done using:

External sensors: GPS, LIDAR, Camera etc.

Internal sensors: IMU, wheel encoders etc.

Simplest, commonly used solution: Odometry

ODOMETRY

Control inputs : v and ω

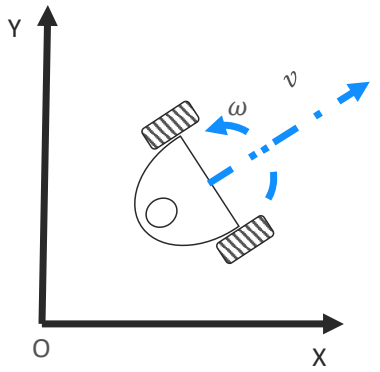
States of robot are x, y position and θ orientation

Kinematic model of diff. drive robot

$$\dot{x} = v\cos\theta$$

$$\dot{y} = v\,sin\,\theta$$

$$\dot{\theta} = \omega$$



Change in position of robot is given by:

$$\Delta x = v\cos\left(\theta + {}^{\Delta\theta}/_{2}\right)\Delta t$$

$$\Delta y = v\,sin\left(\theta + {}^{\Delta\theta}/_{2}\right)\Delta t$$

$$\Delta\theta = \omega\Delta t$$

ODOMETRY

Relationship between control inputs and wheel velocities:

$$v = \frac{(\omega_R + \omega_L)R}{2}$$

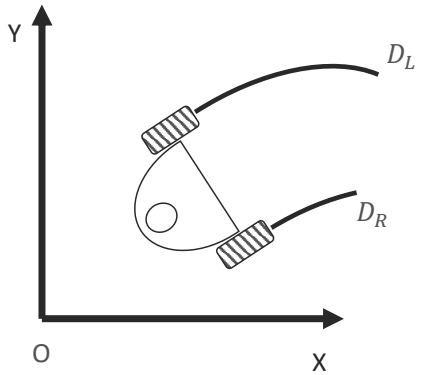
$$\omega = \frac{(\omega_R - \omega_L)R}{L}$$

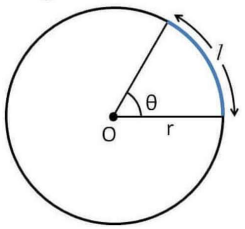
Assume the distance travelled by each wheel $\mathrm{D_L}, \mathrm{D_R}$ are known

Assuming v and ω are constant (valid for short time step, Δt)

→ the robot moves along an arc (general case)

From arc length calculation, $\mathrm{D} = \omega_{L,R} \Delta t \; R$





$l = r\theta$

ODOMETRY

Assuming, distance travelled by left and right

wheel is given by:

$$D_L = R\omega_L \Delta t \text{ and } D_R = R\omega_R \Delta t$$

New pose, P'

$= P +$ change in pose as a function of D_R, D_L



New position of robot is given by:

$$x' = x + \frac{D_R + D_L}{2}\cos(\theta + \frac{D_R - D_L}{2L})$$

$$y' = y + \frac{D_R + D_L}{2}\sin(\theta + \frac{D_R - D_L}{2L})$$

$$\theta' = \theta + \frac{D_R - D_L}{L}$$

ODOMETRY

New position of robot is given by:

$$P' = f(x, y, \theta, D_R, D_L) = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} + \begin{bmatrix} \frac{D_R + D_L}{2} \cos(\theta + \frac{D_R - D_L}{2L}) \\ \frac{D_R + D_L}{2} \sin(\theta + \frac{D_R - D_L}{2L}) \\ \frac{D_R - D_L}{L} \end{bmatrix}$$

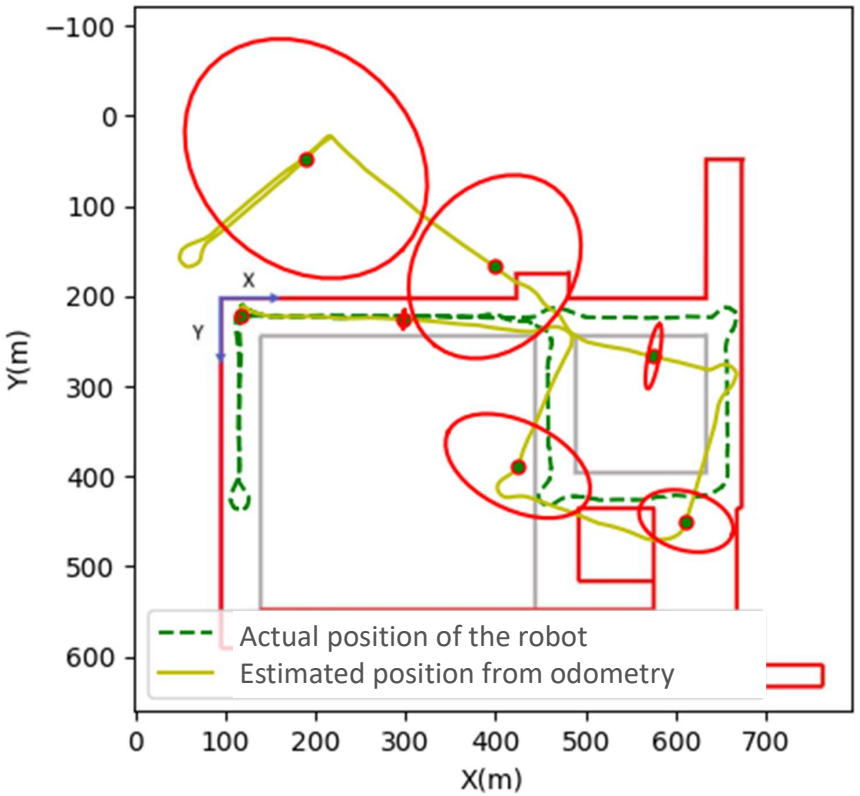
In practice, distance moved by each wheel is given by,

$$D = \frac{\Delta ticks}{(Number\ of\ ticks\ per\ revolution)} \frac{1}{Gear\ Ratio} 2\pi(Radius\ of\ wheel)$$

ODOMETRY

Robot's estimate versus real position of the robot

- Robot estimate is a curved path even when it goes on a straight line
- Robot estimate of turns either undershoots or overshoots



ODOMETRY

Assumptions violated ➔ Sources of error:

× Single point of contact between the wheel and ground plane (Non deformable wheels)

➔ L is no longer deterministic

➔ Uncertainty in L leads to wrong estimates, especially when robot turns

× No sliding/slipping at the point of contact

× No lateral slip/skid

➔Longitudinal and lateral drift in estimated position

ODOMETRY

Assumptions violated → sources of error:

- Misalignment of wheels
- The radius of left and right wheels are not accurate ($\pm$)
- For a given timestep robot moves along constant arc (v, ω constant)
- Slope of floor and external forces acting on robot, collision with moving obstacle

Integration of pose between time steps leads to growing (eventually unbounded) error

Solution:

Use an array of sensors and fuse the pose estimates based on the level of uncertainty associated with each

SENSOR MODELLING

Sources of sensor errors:

Sensors greatly affected by environmental conditions

Cross-sensitivity and robot-environment dynamics introduce error



Example:

- CCD color camera : Effect of change in illumination, robot motion, object motion, reflective material
- Ultrasonic range sensor : Effect of materials, angle at which sound hits obstacle

While these can be modeled and predicted (in theory) they are often not for most field robotic applications

Instead, the errors encountered in mobile robot operation can be assumed random

→ Allowing them to be modelled as stochastic or probabilistic

SENSOR MODELLING

Solution:

Model sensor readings as random variables

$\mathrm{f}(x) = f(X = x)$: Probability that the random variable X has value x

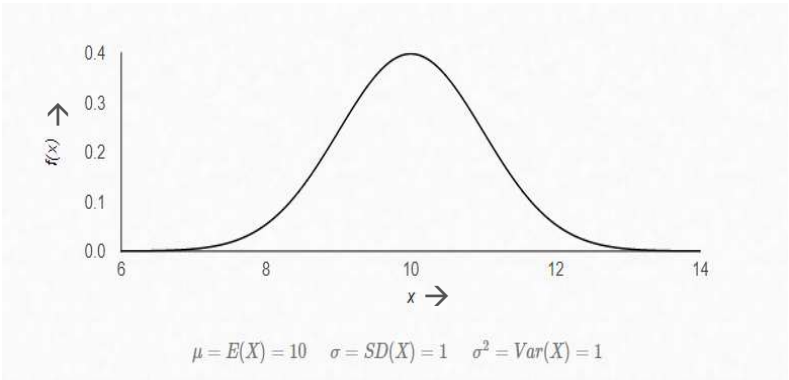
- X : the random variable
- x : a specific value that X can assume

A probability density function (PDF) : describe the likelihood for random variable to take on the given value
Commonly used example: Gaussian distribution

Probability density function for sensor reading x,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

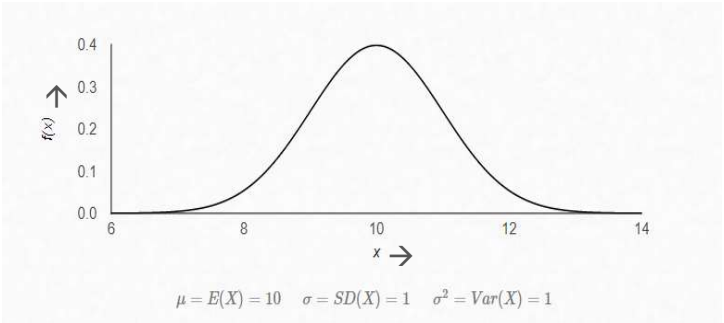
$$= N(x, \sigma^2)$$



SENSOR MODELLING

Probability density function for sensor reading x,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$



Two parameters for representing sensor output:

mean (μ) : The estimated true value for the measurement (expected value of X) is given by the mean value:

$$\mu = \int_{-\infty}^{\infty} xf(x)dx$$

variance (σ^2) : The variance of measurement is given by

$$\sigma^2 = \int_{-\infty}^{\infty}(x-\mu)^2 f(x)dx$$

σ is usually a characteristic of the sensor

SENSOR MODELLING

Example: Model uncertainty associated with an encoder

Two parameters for representing sensor output, mean (μ) and variance (σ^2)

If the encoder reports distance travelled by a wheel as D, then;

$$\mu = D, \sigma^2 = ??$$

SENSOR MODELLING

Example: Model uncertainty associated with an encoder

Two parameters for representing sensor output, mean (μ) and variance (σ^2)

If the encoder reports distance travelled by a wheel as D, then;

$$\mu = D, \sigma^2 = k|D|$$

where, k is a constant (Can be experimentally obtained)

→ the variance is proportional to absolute value of distance travelled by wheel

SENSOR MODELLING

Multivariate case:

When x is a k dimensional vector, the result is a multivariate distribution characterized by a density function,

$$p(x) = \frac{1}{(2\pi)^{\frac{k}{2}} \det(\Sigma)^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

where,

μ is the mean vector and Σ is a positive semidefinite and symmetric matrix called covariance matrix

SENSOR MODELLING

Example: Model uncertainty associated with both encoders

For the two wheels together,

Mean vector is given by

$$\mu = \begin{bmatrix} D_R \\ D_L \end{bmatrix}$$

Covariance matrix is given by;

$$\Sigma_\Delta = \text{covar}(\text{D}_\text{R}, \text{D}_\text{L}) = \begin{bmatrix} k|D_R| & 0 \\ 0 & k|D_L| \end{bmatrix}$$

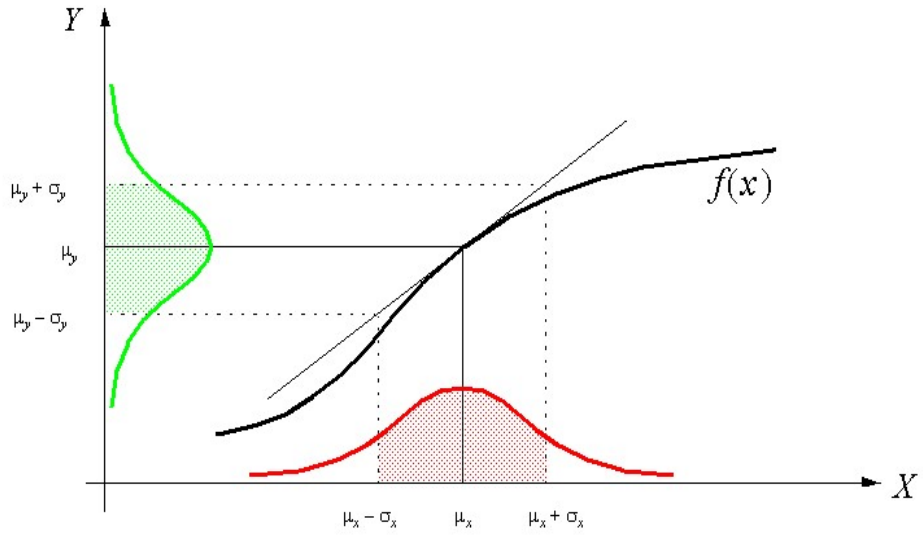
We assume that the two errors from individually driven wheels are independent

Question:

- How do we use fixed uncertainty of encoder to infer uncertainty in the derived robot pose?

ERROR PROPAGATION LAW

One-dimensional case of error propagation:



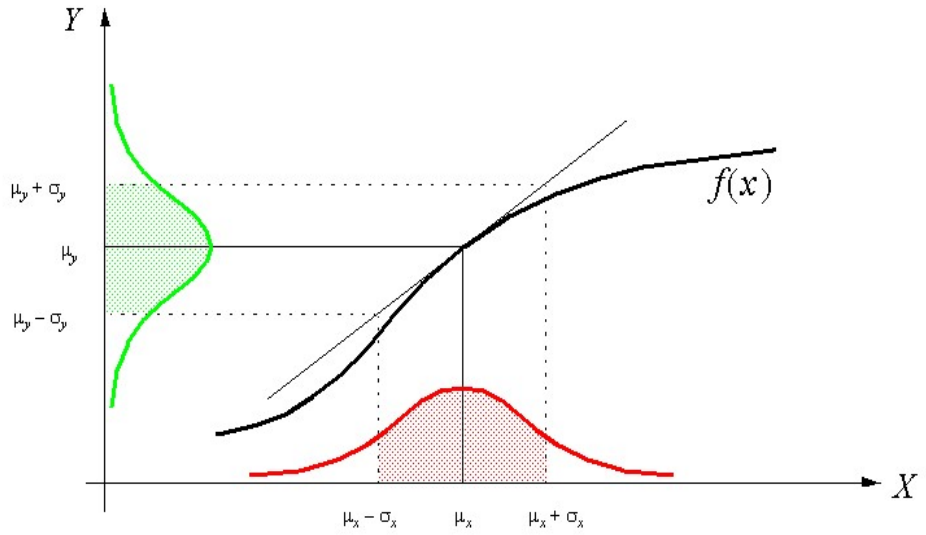
Consider, $y = f(x)$

Given, $x(\mu_x, \sigma_x)$,

calculate μ_y, σ_y

ERROR PROPAGATION LAW

One-dimensional case of error propagation:



Consider, $y = f(x)$

Given, $x(\mu_x, \sigma_x)$,

calculate μ_y, σ_y

$$\mu_y = f(\mu_x)$$

$$\sigma_y^2 = \frac{df}{dx}^2 |_{x=\mu_x} \sigma_x^2$$

ERROR PROPAGATION LAW

Multi-dimensional case of error propagation:

The output covariance matrix C_y is given by the error propagation law:

$$\Sigma_{\mathrm{y}} = F_x \Sigma_x F_x^T$$

where

$Y\ is\ m \times 1$

$\mathrm{X}\ is\ n\ \times 1$

Σ_x : covariance matrix representing the input uncertainties _____

Σ_{y} : covariance matrix representing the propagated uncertainties for the outputs _____

F_x : is the Jacobian matrix_____

ERROR PROPAGATION LAW

Multi-dimensional case of error propagation:

The output covariance matrix C_y is given by the error propagation law:

$$\Sigma_{\mathrm{y}} = F_x \Sigma_x F_x^T$$

where

$Y\ is\ m \times 1$

$\mathrm{X}\ is\ n\ \times 1$

Σ_x : covariance matrix representing the input uncertainties $(n\ \times\ n)$

Σ_{y} : covariance matrix representing the propagated uncertainties for the outputs $(m\ \times m)$

F_x : is the Jacobian matrix $(m\ \times n)$ defined as:

$$F_x = \Delta f = \begin{bmatrix} \frac{\partial f_1}{\partial X_1} & \cdots & \frac{\partial f_1}{\partial X_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial X_1} & \cdots & \frac{\partial f_m}{\partial X_n} \end{bmatrix}$$

ERROR PROPAGATION LAW

Example: Error propagation in odometry

Starting from:

$$P' = f(x, y, \theta, D_R, D_L) = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} + \begin{bmatrix} \frac{D_R + D_L}{2} \cos(\theta + \frac{D_R - D_L}{2L}) \\ \frac{D_R + D_L}{2} \sin\left(\theta + \frac{D_R - D_L}{2L}\right) \\ \frac{D_R - D_L}{L} \end{bmatrix}$$

Given,

$$\Sigma_\Delta = \text{covar}(\mathrm{D_R}, \mathrm{D_L}) = \begin{bmatrix} k|D_R| & 0 \\ 0 & k|D_L| \end{bmatrix} \text{ (2x2 matrix)}$$

$$\Sigma_{P'} = F_\Delta \, \Sigma_\Delta F_\Delta^T$$

ERROR PROPAGATION LAW

Example: Error propagation in odometry

$$F_\Delta = \begin{bmatrix} \frac{\partial f}{\partial D_R} & \frac{\partial f}{\partial D_L} \end{bmatrix} \text{ (3x2 matrix)}$$

$$= \begin{bmatrix} & \\ & \\ & \end{bmatrix}$$

ERROR PROPAGATION LAW

Example: Error propagation in odometry

$$F_\Delta = \begin{bmatrix} \frac{\partial f}{\partial D_R} & \frac{\partial f}{\partial D_L} \end{bmatrix} \text{ (3x2 matrix)}$$

$$= \begin{bmatrix} \frac{1}{2}\cos\left(\theta + \frac{\Delta\theta}{2}\right) - \frac{\Delta D}{2L}\sin(\theta + \frac{\Delta\theta}{2}) & \frac{1}{2}\cos\left(\theta + \frac{\Delta\theta}{2}\right) + \frac{\Delta D}{2L}\sin(\theta + \frac{\Delta\theta}{2}) \\ \frac{1}{2}\sin\left(\theta + \frac{\Delta\theta}{2}\right) + \frac{\Delta D}{2L}cos(\theta + \frac{\Delta\theta}{2}) & \frac{1}{2}\sin\left(\theta + \frac{\Delta\theta}{2}\right) - \frac{\Delta D}{2L}cos(\theta + \frac{\Delta\theta}{2}) \\ \frac{1}{L} & -\frac{1}{L} \end{bmatrix}$$

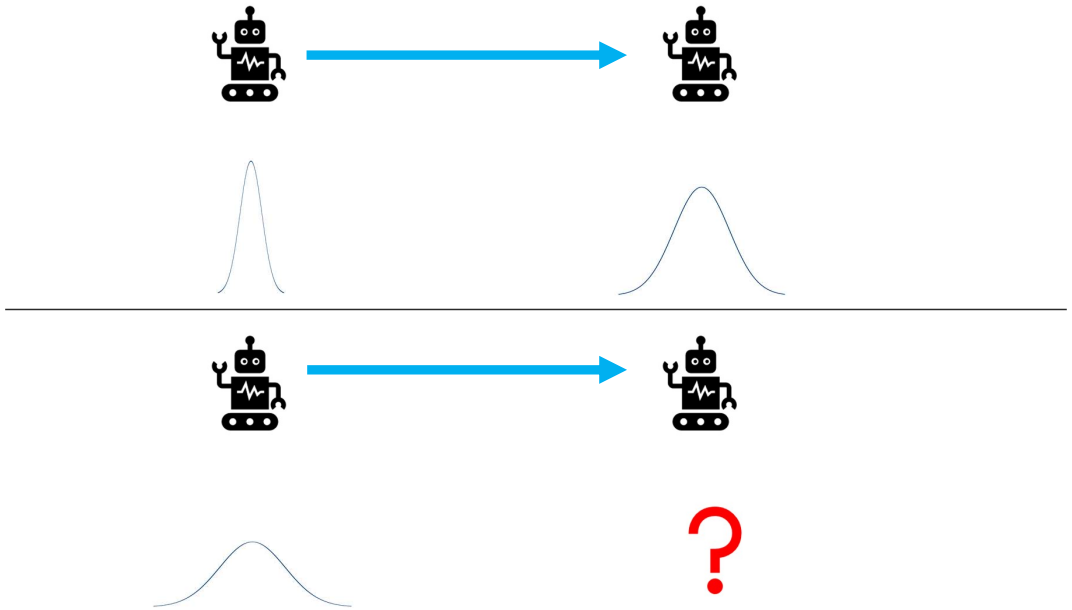
where,

$$\Delta D = \frac{\mathrm{D_R} + \mathrm{D_L}}{2}, \Delta\theta = \frac{D_R - D_L}{L}$$

ERROR PROPAGATION LAW

Question:

- Does the uncertainty in the derived robot pose depend only on the distance moved by the robot ?



ERROR PROPAGATION LAW

Example: Error propagation in odometry

Taking into account uncertainty in current robot pose

$$\Sigma_{P'} = F_P \Sigma_P F_P^T + F_\Delta \Sigma_\Delta F_\Delta^T$$

Σ_P (3x3 matrix) can be initialized to zero matrix in beginning and then updated over subsequent steps

$$F_\mathrm{P} = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial \theta} \end{bmatrix}$$ (__ matrix)

$$= \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$$

ERROR PROPAGATION LAW

Example: Error propagation in odometry

Taking into account uncertainty in current robot pose

$$\Sigma_{P'} = F_P \Sigma_P F_P^T + F_\Delta \Sigma_\Delta F_\Delta^T$$

Σ_P (3x3 matrix) can be initialized to zero matrix in beginning and then updated over subsequent steps

$$F_\mathrm{P} = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial \theta} \end{bmatrix}$$ (3x3 matrix)

$$= \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$$

ERROR PROPAGATION LAW

Example: Error propagation in odometry

Taking into account uncertainty in current robot pose

$$\Sigma_{P'} = F_P \Sigma_P F_P^T + F_\Delta \Sigma_\Delta F_\Delta^T$$

Σ_P (3x3 matrix) can be initialized to zero matrix in beginning and then updated over subsequent steps

$$F_{\mathrm{P}} = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial \theta} \end{bmatrix} \text{ (3x3 matrix)}$$

$$= \begin{bmatrix} 1 & 0 & -\Delta D \sin(\theta + \frac{\Delta\theta}{2}) \\ 0 & 1 & \Delta D \cos(\theta + \frac{\Delta\theta}{2}) \\ 0 & 0 & 1 \end{bmatrix}$$

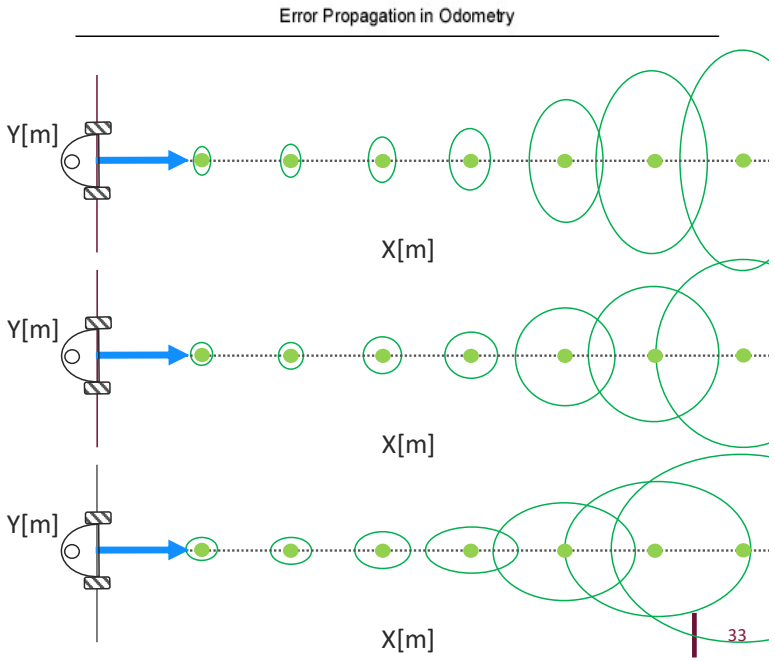
where,

$$\Delta D = \frac{\mathrm{D_R} + \mathrm{D_L}}{2}, \Delta\theta = \frac{D_R - D_L}{L}$$

ERROR PROPAGATION LAW

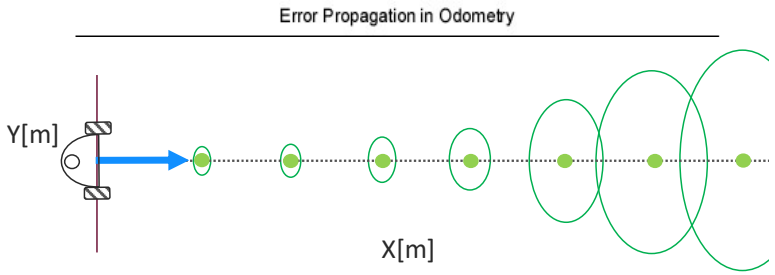
Question: Which figure accurately represents growth of

pose uncertainty for straight line motion



ERROR PROPAGATION LAW

Question: Which figure accurately represents growth of pose uncertainty for straight line motion



Uncertainty in y (local) grows much faster than in the direction of movement.

This results from the integration of uncertainty about robot's orientation.

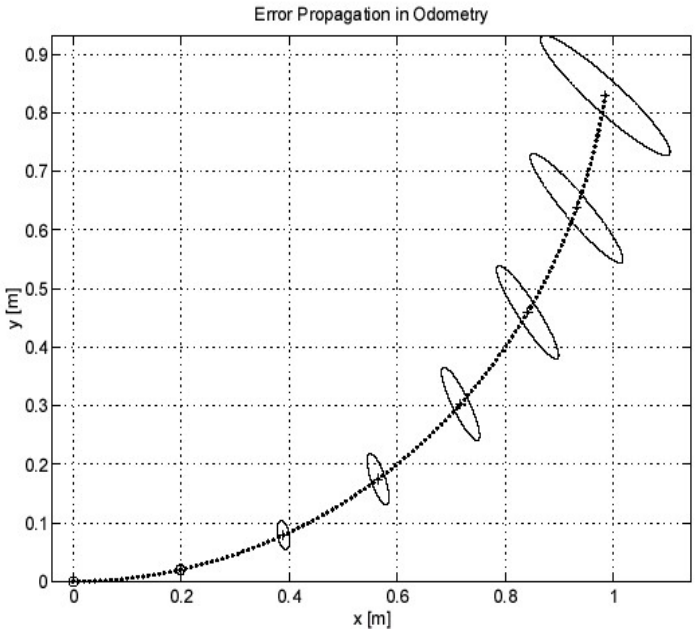
The ellipses drawn around the robot positions represent the uncertainties in xy direction.

ERROR PROPAGATION LAW

Growth of pose uncertainty for circular motion

Uncertainty grows faster perpendicular to the directior of motion.

Error ellipse does not remain perpendicular to the direction of motion



Question:

- Can we make use of other sensor readings to improve localization ?

MAP-BASED LOCALIZATION

Extra reading:

- Chapter 5 of “Introduction to Autonomous Mobile robots” book

  Sections 5.1, 5.2

BASICS OF PROBABILITY

- $p(x, z)$: Joint distribution representing the probability that the random variable X takes on x and Z takes on z:

$$p(x, z) = p(x)p(z)$$

provided X and Z are independent

- $p(x|z)$: Conditional probability describes the probability that the random variable X takes on x, under the condition that Z for sure takes on z:

$$p(x|z) = \frac{p(x, z)}{p(z)}$$

BASICS OF PROBABILITY

Theorem of total probability:

$$p(x) = \int_z p(x|z)p(z)dz$$

(Σ for discrete probabilities)

Question: Coin flip game

Given a fair coin (0.5 chance of Head or Tail), game rules as follows:

Toss the coin and if we get Tail, we accept the result. If we get Head, we toss again and accept whatever is the result the second time.

Given the above rule compute probability of getting Head for the second toss?

BAYES RULE

Bayes rule relates the conditional probability of $p(x|z)$ to its inverse $p(z|x)$:

$$p(x|z) = \frac{p(z|x)p(x)}{p(z)}$$

provided $p(z) > 0$

$$p(x|z) = \eta p(z|x)p(x)$$

where,

$\eta = p(z)^{-1}$, known as the normalization factor which ensures $\int p\,(x) = 1$

BASICS OF PROBABILITY

Deriving Baye's rule from joint distribution and conditional probability definitions

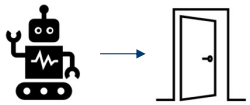
$$p(x|z) = \frac{p(x,z)}{p(z)} \text{ and } p(z|x) = \frac{p(z,x)}{p(x)}$$

$$\rightarrow p(x|z)p(z) = p(z|x)p(x)$$

$$p(x|z) = \frac{p(z|x)p(x)}{p(z)}$$

BAYES RULE

Question: Door test



Probability of having a door being open is given by $p(d) = 0.001$

Let's say there is a sensor that can detect positive when the door is open. The sensor statistics are given by:

Probability of sensor detecting positive when door is open, $p(\ +ve \mid d\ ) = 0.8$

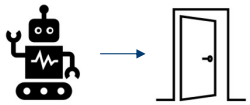
Probability of sensor detecting positive when door is closed, $p(\ +ve \mid \sim d\ ) = 0.1$

PS: notice that the above two does not sum up to zero, why?

Compute, probability of having the door open when the sensor detects +ve, $p(\ d \mid +\ ve) = ?$

BAYES RULE

Question: Door test



Probability of having a door being open is given by $p(d) = 0.5$

Let's say there is a sensor that can detect positive when the door is open. The sensor statistics are given by:

Probability of sensor detecting positive when door is open, $p(\ +ve \mid d\ ) = 0.8$

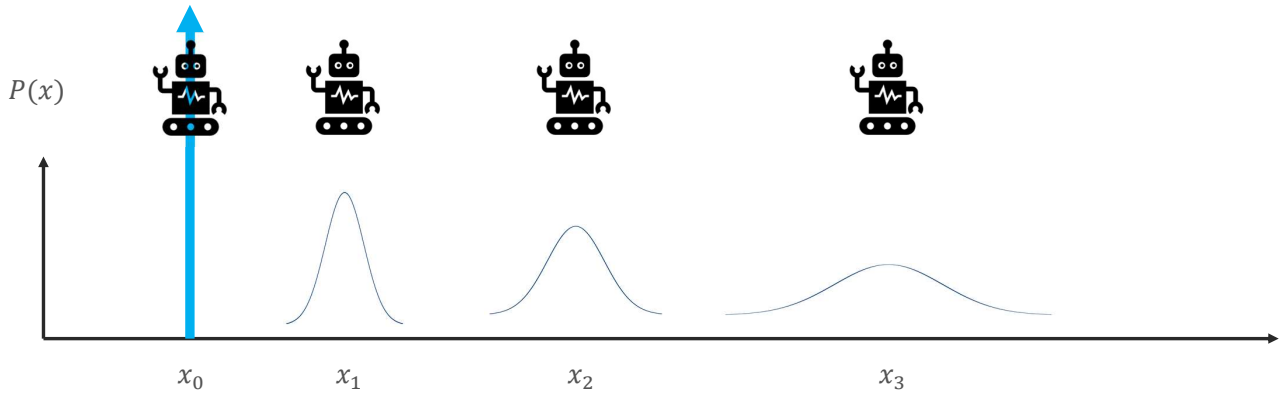
Probability of sensor detecting positive when door is closed, $p(\ +ve \mid \sim d\ ) = 0.1$

PS: notice that the above two does not sum up to zero, why?

Compute, probability of having the door open when the sensor detects +ve, $p(\ d \mid +\ ve) = ?$

BAYES RULE

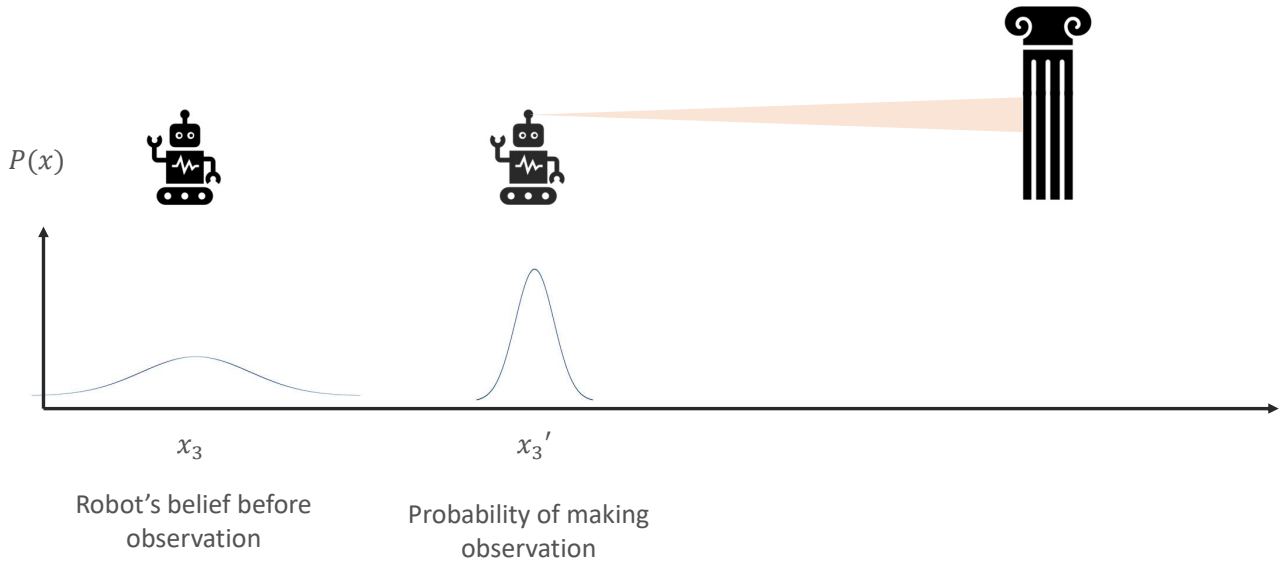
Robot starts at x_0, fully certain, but as it moves state uncertainty obtained through odometry grows



BAYES RULE

Robot makes an observation using a range sensor of a known landmark

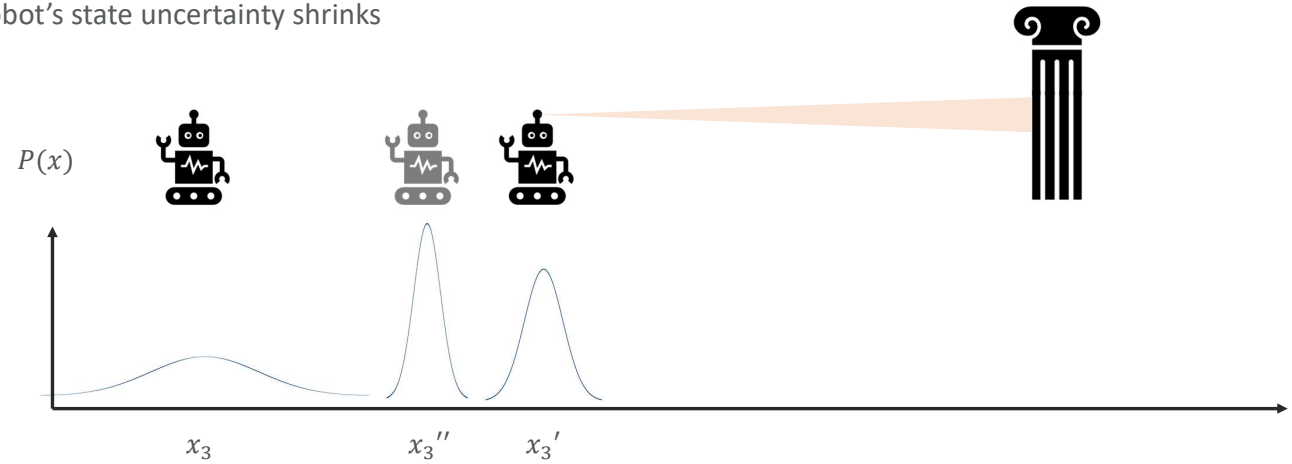
→ Another estimate of current position



BAYES RULE

Robot can correct its position by combining its belief before observation with the probability of making the observation

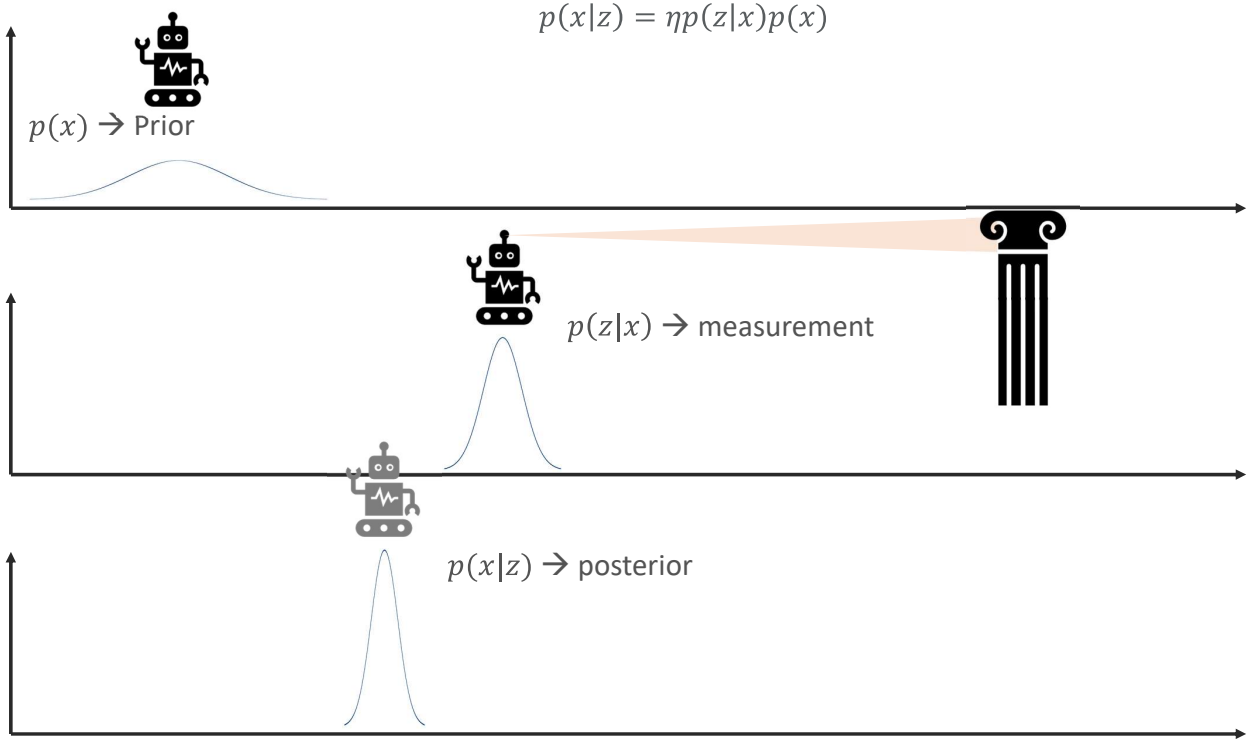
→ Robot's state uncertainty shrinks



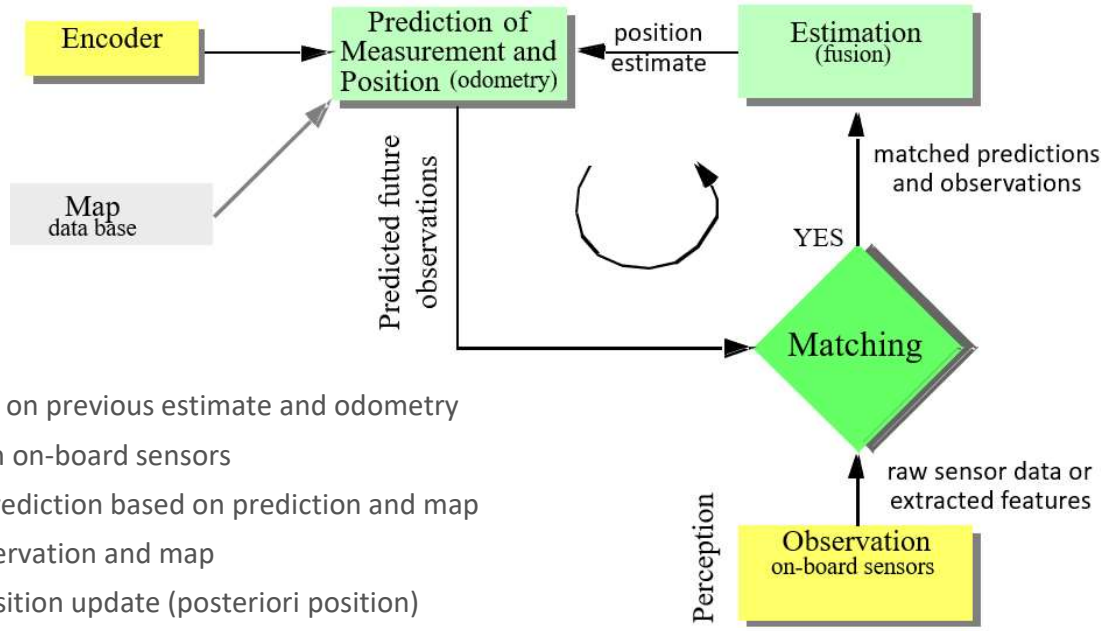
Robot belief update

BAYES RULE

$$p(x|z) = \eta p(z|x)p(x)$$



MAP-BASED LOCALIZATION



1. Prediction based on previous estimate and odometry
2. Observation with on-board sensors
3. Measurement prediction based on prediction and map
4. Matching of observation and map
5. Estimation -> position update (posteriori position)

MAP-BASED LOCALIZATION

Extra reading:

- Chapter 5 of “Introduction to Autonomous Mobile robots” book

  Sections 5.6.1 – 5.6.6

MAP-BASED LOCALIZATION

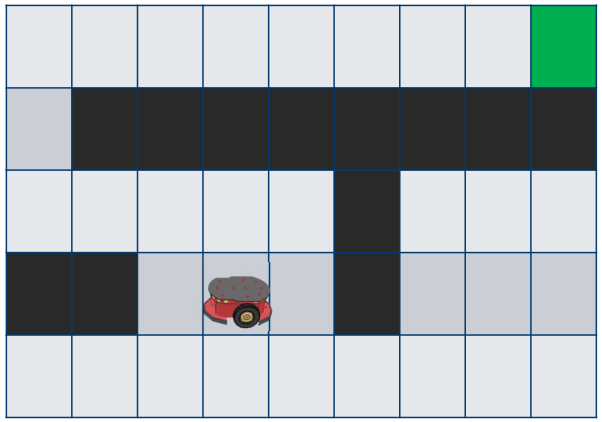
Map based localization approaches:

1. Markov localization
2. Kalman filter localization

MARKOV LOCALIZATION

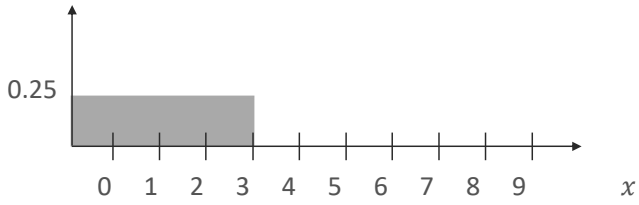
Markov localization

1. Uses a discretized representation of robot pose, such as a grid map
2. The robot's position can be an arbitrary probability density functions –> not necessarily a Gaussian
3. Assumption: New state of robot (x_t) is only a function of previous state (x_{t-1}), action (u_t), and perception (z_t)
4. Markov localization addresses position tracking, global localization problem, and kidnapped robot problem

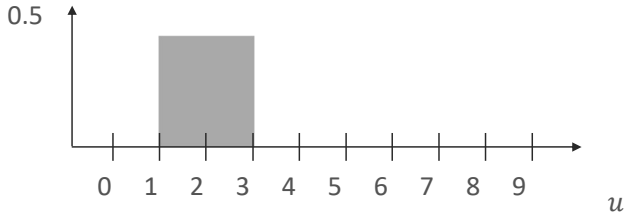


MARKOV LOCALIZATION

Localization about a static grid map



$bel(x_0)$



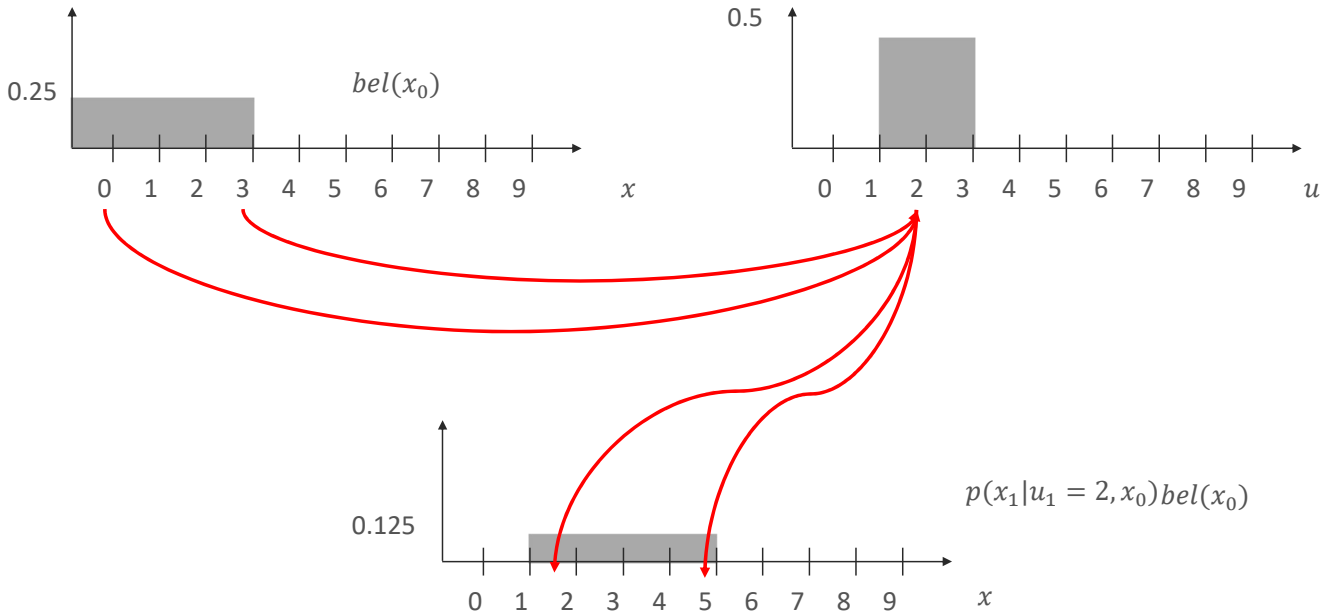
$p(u_1)$

?

$\overline{bel}(x_1)$

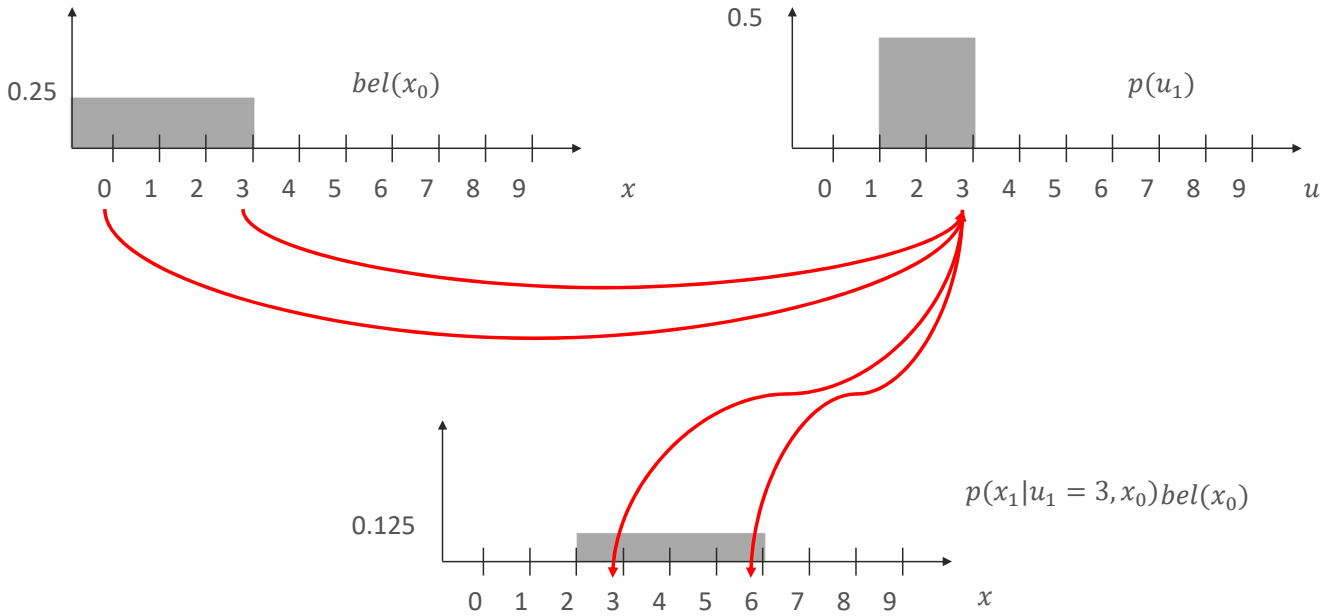
MARKOV LOCALIZATION

Localization about a static grid map : Prediction using theorem of Total Probability / convolution operation



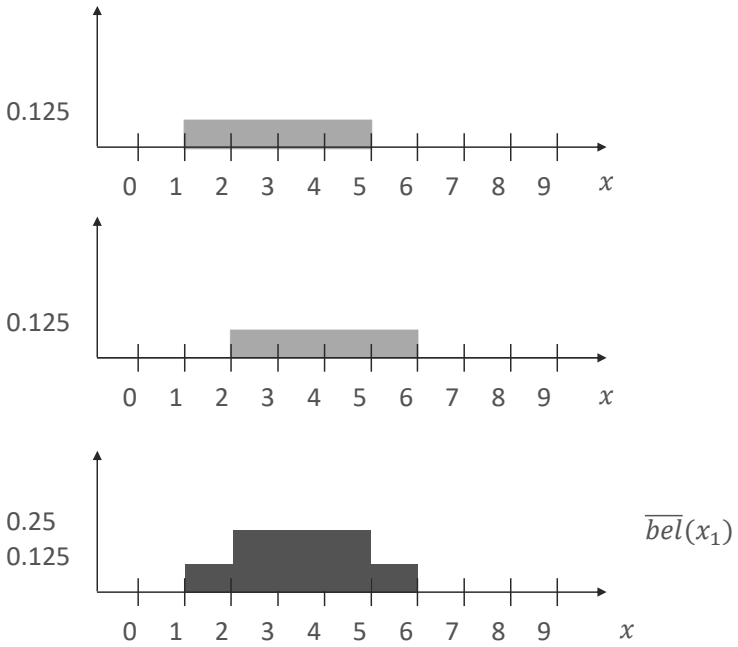
MARKOV LOCALIZATION

Localization about a static grid map : Prediction using theorem of Total Probability / convolution operation



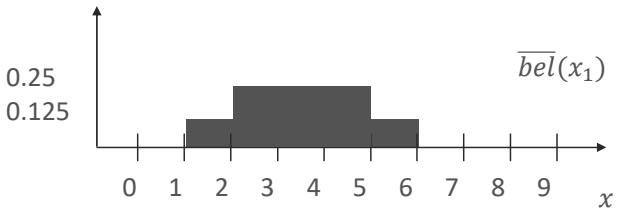
MARKOV LOCALIZATION

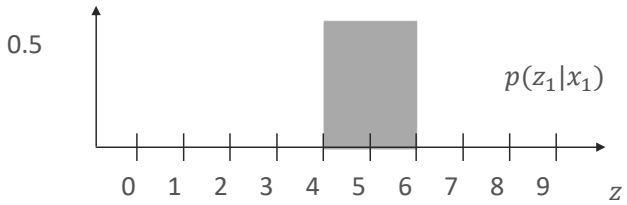
Localization about a static grid map



MARKOV LOCALIZATION

Localization about a static grid map: correction using Bayes rule



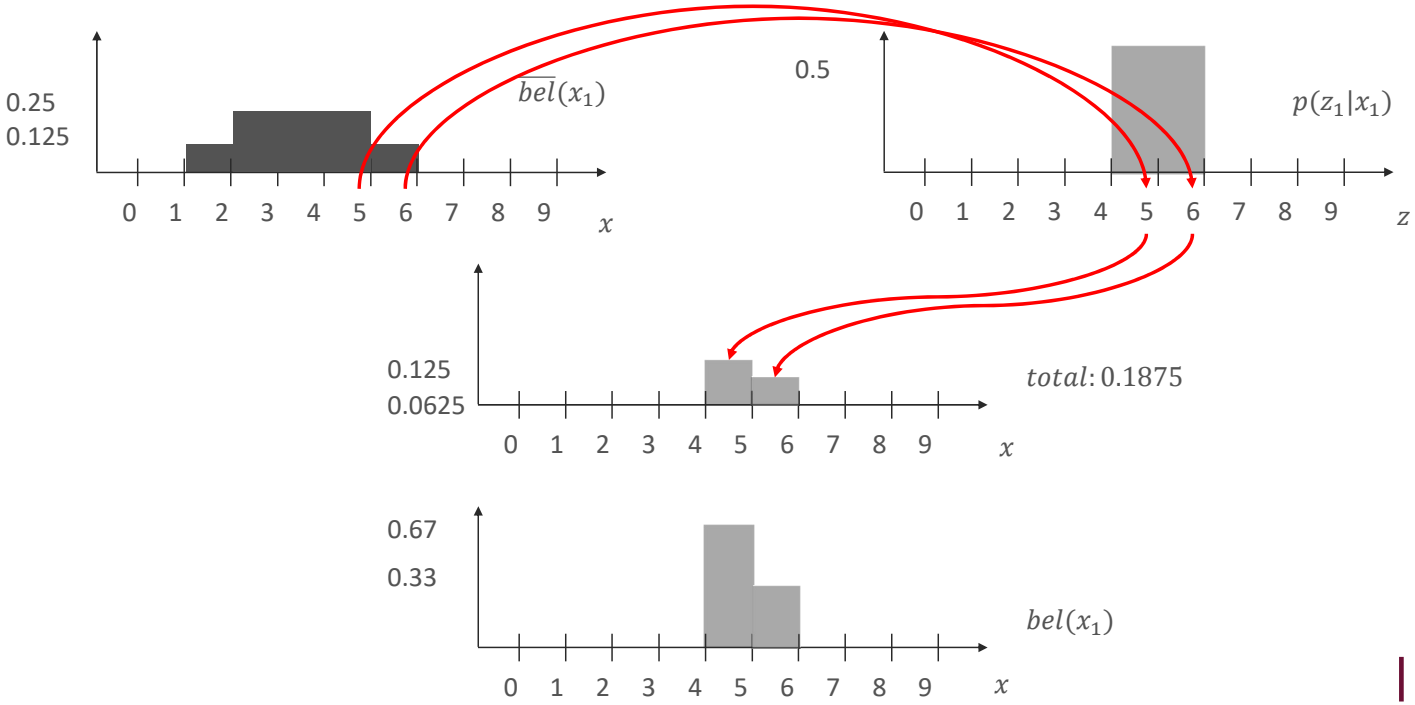


?

$bel(x_1)$

MARKOV LOCALIZATION

Localization about a static grid map: correction using Bayes rule



MARKOV LOCALIZATION

Algorithm:

For all x_t, do:

Prediction update

$$\overline{bel}(x_t) = \sum_{x_{t-1}} p(x_t|u_t, x_{t-1}) bel(x_{t-1}) dx_{t-1}$$

Measurement update

$$bel(x_t) = \eta p(z_t|x_t) \overline{bel}(x_t)$$

MARKOV LOCALIZATION

Advantages:

- Localization starting from any unknown position → Kidnapped robot problem
- Recovers from ambiguous situation.

Disadvantage: Not very efficient:

- Discrete representation of the space (grid)
- Probability updates of all positions within the whole state space at any time
- The required memory and calculation power becomes too much if a fine grid / higher dimensional space is to be considered (example quadcopter in open space)

MAP-BASED LOCALIZATION

Extra reading:

- Chapter 5 of “Introduction to Autonomous Mobile robots” book

  Sections 5.6.7 Markov localization (Section 5.6.2 in older versions)

KALMAN FILTER LOCALIZATION

A mathematical mechanism for producing an optimal estimate of system state based on the knowledge of:

- the system (models, system noise, uncertainty in the dynamic models)
- the measuring device (measurement errors/noise)

Kalman filter

- Operates on a continuous representation of states
- A powerful method for sensor fusion
- Can be used for Localization about a static map for a linear system, with Gaussian noise
    - → Next state is a linear function of current state and control applied
    - → Observation/measurement is a linear function of state

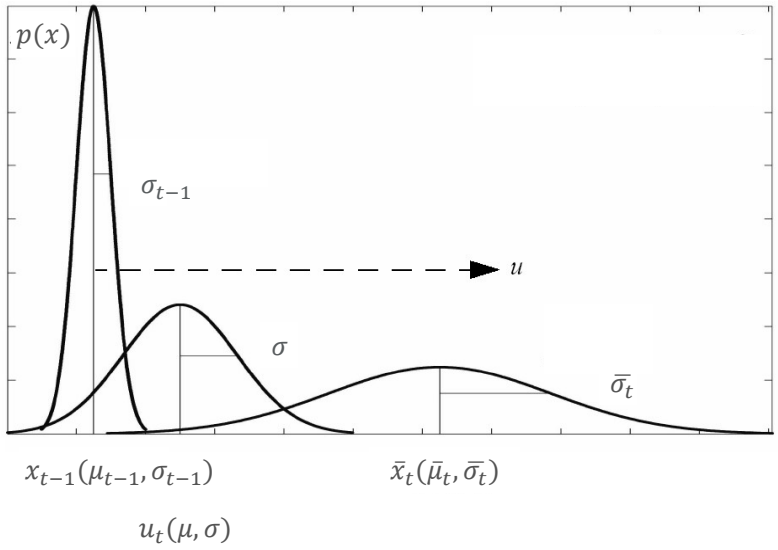
KALMAN FILTER LOCALIZATION

Given: $x_{t-1}(\mu_{t-1}, \sigma_{t-1})$ and $u_t(\mu, \sigma)$

for calculating, $\overline{x}_t(\bar{\mu}_t, \overline{\sigma}_t)$ we can apply theorem of total probability :

Assuming, $x_t = x_{t-1} + u_t$

$$\bar{\mu}_t = \mu_{t-1} + \mu$$
$$\overline{\sigma}_t^2 = \sigma_{t-1}^2 + \sigma^2$$



This is called the prediction step for a Kalman filter

THEOREM OF TOTAL PROBABILITY

Question: Theorem of total probability rule for two gaussians

Assume initial location of the robot is given by a Gaussian distribution,

$$\mu_1 = 8, \sigma_1^2 = 4$$

Assume the motion of the robot is also represented by a Gaussian distribution,

$$\mu_2 = 2, \sigma_2^2 = 2$$

Find a resulting Gaussian distribution that represents the location of the robot after executing the motion?

KALMAN FILTER LOCALIZATION

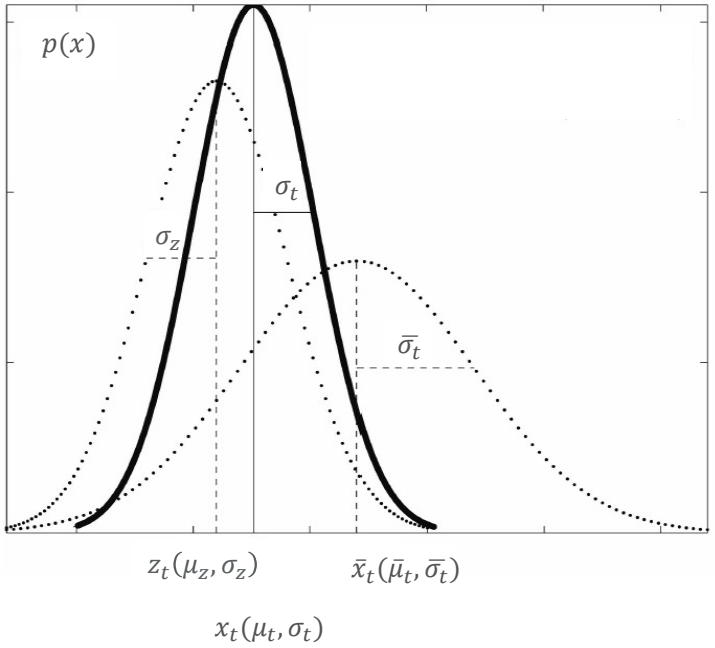
Given: $\bar{x}_t(\bar{\mu}_t, \bar{\sigma}_t)$ and measurement $z_t(\mu_z, \sigma_z)$

for calculating, $x_t(\mu_t, \sigma_t)$ we can apply Bayes rule:

$$\mu_t = \frac{\frac{\bar{\mu}_t}{\bar{\sigma}_t^2} + \frac{\mu_z}{\sigma_z^2}}{\frac{1}{\bar{\sigma}_t^2} + \frac{1}{\sigma_z^2}}$$

$$\frac{1}{\sigma_t^2} = \frac{1}{\bar{\sigma}_t^2} + \frac{1}{\sigma_z^2}$$

This is called the correction step for a Kalman filter



KALMAN FILTER LOCALIZATION

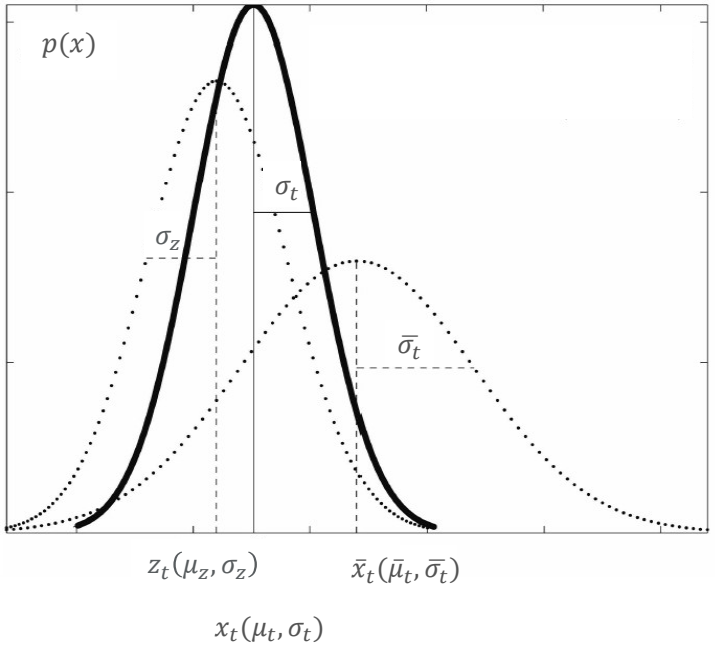
Given: $\bar{x}_t(\bar{\mu}_t, \bar{\sigma}_t)$ and measurement $z_t(\mu_z, \sigma_z)$

for calculating, $x_t(\mu_t, \sigma_t)$ we can apply Bayes rule:

$$\mu_t = \frac{\sigma_z^2 \bar{\mu}_t + \bar{\sigma}_t^{\,2} \mu_z}{\bar{\sigma}_t^{\,2} + \sigma_z^2}$$

$$\frac{1}{\sigma_t^2} = \frac{1}{\bar{\sigma}_t^{\,2}} + \frac{1}{\sigma_z^2}$$

This is called the correction step for a Kalman filter



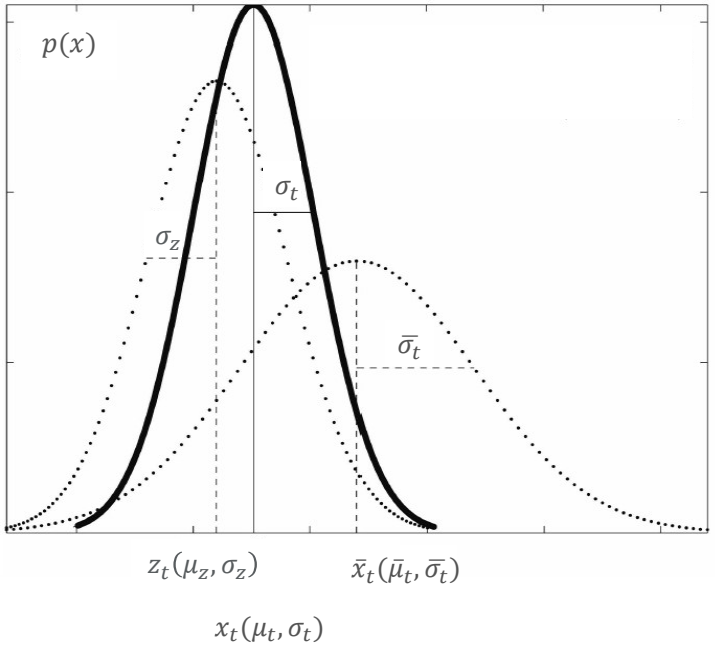
KALMAN FILTER LOCALIZATION

The correction equations can also be written as:

$$\mu_t = \bar{\mu}_t + \frac{\bar{\sigma}_t^{\,2}}{\bar{\sigma}_t^{\,2} + \sigma_z^2}(\mu_z - \bar{\mu}_t)$$

$$\sigma_t^2 = \bar{\sigma}_t^{\,2} - \frac{\bar{\sigma}_t^{\,2}}{\bar{\sigma}_t^{\,2} + \sigma_z^2}(\bar{\sigma}_t^{\,2})$$

The term $\frac{\bar{\sigma}_t^{\,2}}{\bar{\sigma}_t^{\,2}+\sigma_z^2}$ is often referred to as Kalman gain



BAYES RULE

Question: Bayes rule for two gaussians

Given two Gaussians distributions that represent the same quantity, with mean and variance as:

$$\mu_1 = 10, \sigma_1^2 = 4$$

$$\mu_2 = 12, \sigma_2^2 = 4$$

Find a resulting combined gaussian distribution using Bayes rule?

KALMAN FILTER LOCALIZATION

Special case of Markov localization:

Uses Gaussians to represent belief, motion and measurement models

→ Inherently very precise

→ Very efficient : Only two parameters to be updated and computed, mean μ_t and covariance Σ_t

→ Most of computational load comes from matrix inversion

Limitation:

Initial belief $bel(x_0)$ also must be a gaussian

Cannot start from uniform distribution as in case of Markov localization

Can only be used for tracking with initial guess, not for global localization or kidnapped robot problems

If the uncertainty of the robot becomes too large (e.g., collision with an object) the Kalman filter will fail

State transitions and measurements are rarely linear in practice

LINEARIZATION

Question: Create a linear approximation for $y = \sin x$

Linear approximation for $f(x)$ is given by:

$$f(x) = f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \cdots$$

Assuming $x_0 = 0$;

$$y = ??$$

LINEARIZATION

Question: Create a linear approximation for $y = \sin x$

$$\sin x = \mathrm{x} - \frac{x^3}{3!} + \frac{x^5}{5!} \ldots$$

Linearize around $x = 0$

Linear approximation is $y = x$

Linearization is accurate close to $x = 0$

and deviates as x moves farther from zero.

How to fix this??

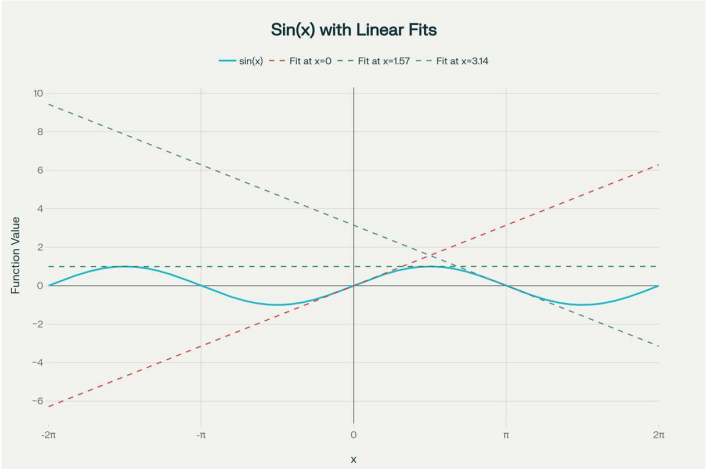


LINEARIZATION

Question: Create a linear approximation for $y = \sin x$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} \ldots$$

At $x = 0$: Linear approximation is $y = x$.
At $x = 1.57$: Linear approximation is $y = 1$.
At $x = 3.14$: Linear approximation is $y = 3.14 - x$.



EXTENDED KALMAN FILTER

Prediction step:

Example: Differential drive robot

$$\text{New pose} = f(x, y, \theta, D_R, D_L) = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} + \begin{bmatrix} \frac{D_R + D_L}{2} \cos(\theta + \frac{D_R - D_L}{2L}) \\ \frac{D_R + D_L}{2} \sin\left(\theta + \frac{D_R - D_L}{2L}\right) \\ \frac{D_R - D_L}{L} \end{bmatrix}$$

$$\hat{X}_t = F(U_t, X_{t-1})$$

where,

$$X = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}, U = \begin{bmatrix} D_R \\ D_L \end{bmatrix}$$

EXTENDED KALMAN FILTER

Prediction step:

Example: Differential drive robot

New covariance matrix, $\widehat{P}_t = F_P P_{t-1} F_P^T + F_u Q_t F_u^T$

where,

$$F_P = \begin{bmatrix} 1 & 0 & -\Delta D \sin(\theta + \frac{\Delta\theta}{2}) \\ 0 & 1 & \Delta D \cos(\theta + \frac{\Delta\theta}{2}) \\ 0 & 0 & 1 \end{bmatrix}, P_{t-1} \text{ is old covariance, } Q_t(\text{motion model covariance}) = \begin{bmatrix} k|D_{R,t}| & 0 \\ 0 & k|D_{L,t}| \end{bmatrix},$$

$$F_u = \begin{bmatrix} \frac{1}{2}\cos\left(\theta + \frac{\Delta\theta}{2}\right) - \frac{\Delta D}{2L}\sin(\theta + \frac{\Delta\theta}{2}) & \frac{1}{2}\cos\left(\theta + \frac{\Delta\theta}{2}\right) + \frac{\Delta D}{2L}\sin(\theta + \frac{\Delta\theta}{2}) \\ \frac{1}{2}\sin\left(\theta + \frac{\Delta\theta}{2}\right) + \frac{\Delta D}{2L}cos(\theta + \frac{\Delta\theta}{2}) & \frac{1}{2}\sin\left(\theta + \frac{\Delta\theta}{2}\right) - \frac{\Delta D}{2L}cos(\theta + \frac{\Delta\theta}{2}) \\ \frac{1}{L} & -\frac{1}{L} \end{bmatrix}$$

EXTENDED KALMAN FILTER

Correction step:

Example: Sensor that measures square of the distance from origin

Observation: Square of distance of robot from origin, z

Covariance of observation, R: Could be a function of distance of robot from origin or constant

Single value in this case, so R is (1x1 matrix)

Measurement prediction $= h(x, y) =$??

EXTENDED KALMAN FILTER

Correction step:

Example: Sensor that measures square of the distance from origin

Observation: Square of distance of robot from origin, z

Covariance of observation, R: Could be a function of distance of robot from origin or constant

Single value in this case, so R is (1x1 matrix)

$$\text{Measurement prediction} = h(x, y) = x^2 + y^2$$

$$\text{Innovation, } v_t = z_t - h(\widehat{X_t})$$

$$\text{Innovation covariance, } \Sigma_{IN\ t} = H\widehat{P}_t H^T + R_t$$

where,

$$H = ??$$

EXTENDED KALMAN FILTER

Correction step:

Example: Sensor that measures square of the distance from origin

Observation: Square of distance of robot from origin, z

Covariance of observation, R: Could be a function of distance of robot from origin or constant

Single value in this case, so R is (1x1 matrix)

$$\text{Measurement prediction} = h(x, y) = x^2 + y^2$$

$$\text{Innovation, } v_t = z_t - h(\widehat{X_t})$$

$$\text{Innovation covariance, } \Sigma_{IN\ t} = H\widehat{P}_t H^T + R_t$$

where,

$$H = [2x \quad 2y \quad 0] \text{ (1x3 matrix)}$$

EXTENDED KALMAN FILTER

Correction step:

Example: Sensor that measures square of the distance from origin

$$\text{Kalman gain, } \mathrm{K_t} = \widehat{P}_t H^T \, (\Sigma_{IN_t})^{-1}$$

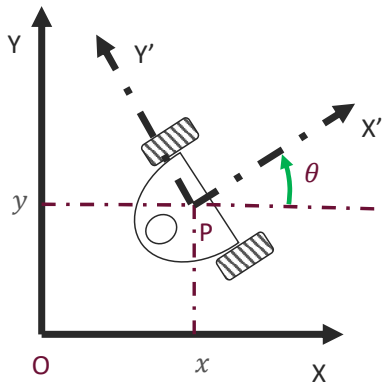
$$\text{Mean, } \mathrm{X_t} = \hat{X}_t + K_t \, v_t$$

The best estimate of robot state at time t, $(\mathrm{X_t})$ is the best prediction value before the new measurement $(\hat{X}_t)$, plus a correction term of an optimal weight $(\mathrm{K_t})$ times the difference between the new measurement (z_t) and the predicted measurement, $h(\widehat{X_t})$.

$$\text{Covariance, } P_t = \widehat{P}_t - \mathrm{K_t}\Sigma_{IN_t}K_t^T$$
$$= (\mathrm{I} - \mathrm{K_t}\mathrm{H_t})\widehat{P}_t$$

EXTENDED KALMAN FILTER

Given, Initial state, $X_0 = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0.5235 \end{bmatrix}$, width of the robot is $0.5m$



Initial covariance, $P_0 = \begin{bmatrix} 0.1 & 0 & 0 \\ 0 & 0.1 & 0 \\ 0 & 0 & 0.017 \end{bmatrix}$

Motion/prediction step

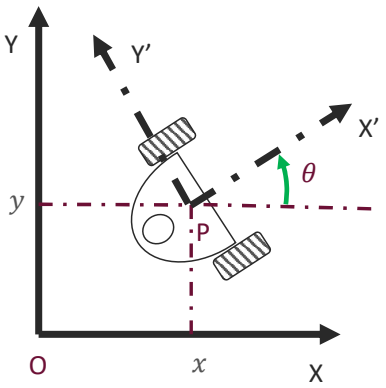
Control input, $U_0 = \begin{bmatrix} D_R \\ D_L \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, Motion covariance, $Q_0 = \begin{bmatrix} 0.2 & 0 \\ 0 & 0.2 \end{bmatrix}$

Compute $\hat{X}_1, \widehat{P_1}$

EXTENDED KALMAN FILTER

Motion/prediction step

$$\hat{X}_1 = \begin{bmatrix} 2.7 \\ 1.9 \\ 0.5235 \end{bmatrix}, \widehat{P_1} = \begin{bmatrix} 0.59 & -0.67 & -0.81 \\ -0.67 & 1.37 & 1.41 \\ -0.81 & 1.41 & 1.67 \end{bmatrix}$$



EXTENDED KALMAN FILTER

<u>Measurement/correction step</u>

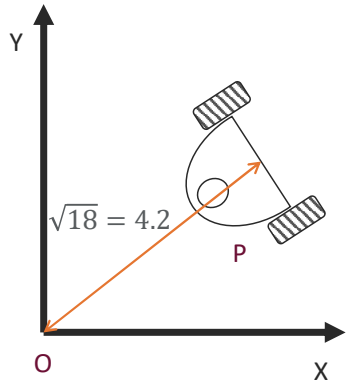
Square of distance of robot from origin, $z_1 = 18$

Covariance of observation, $R = 0.05$

Measurement prediction $= h(\widehat{X_1}) =$

Innovation, $v_1 = z_1 - h(\widehat{X_1}) =$

$$H = [2x \quad 2y \quad 0] =$$



EXTENDED KALMAN FILTER

<u>Measurement/correction step</u>

Square of distance of robot from origin, $z_1 = 18$

Covariance of observation, $R = 0.05$

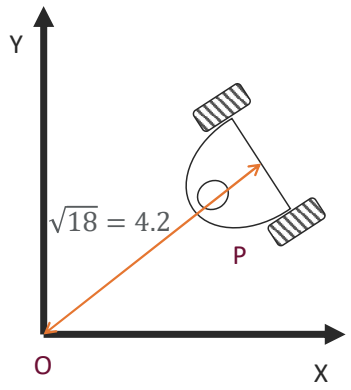
Measurement prediction $= h(\widehat{X_1}) = \hat{x}^2 + \hat{y}^2 = 11.46$

Innovation, $v_1 = z_1 - h(\widehat{X_1})$ = 6.536

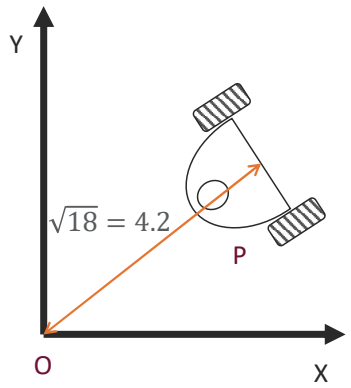
$H = [2x \quad 2y \quad 0] = [5.4 \quad 3.9 \quad 0]$

Innovation covariance, $\Sigma_{IN\,1} = H\widehat{P_1}H^T + R_1 =$

Kalman gain, $\mathrm{K}_1 = \widehat{P_1}H^T\,(\Sigma_{IN_1})^{-1} =$



EXTENDED KALMAN FILTER



Measurement/correction step

Square of distance of robot from origin, $z_1 = 18$

Covariance of observation, $R = 0.05$

Measurement prediction $= h(\widehat{X_1}) = \hat{x}^2 + \hat{y}^2 = 11.46$

Innovation, $v_1 = z_1 - h(\widehat{X_1}) = 6.536$

$H = [2x \quad 2y \quad 0] = [5.4 \quad 3.9 \quad 0]$

Innovation covariance, $\Sigma_{IN\,1} = H\widehat{P_1}H^T + R_1 = 10.0621$

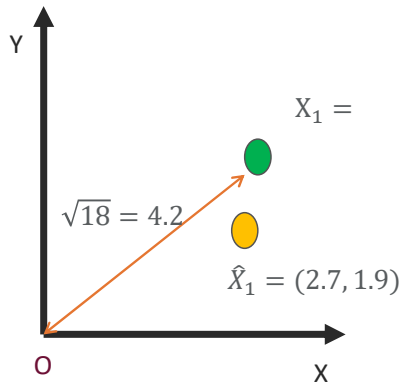
Kalman gain, $\mathrm{K}_1 = \widehat{P_1}H^T\,(\Sigma_{IN_1})^{-1} = \begin{bmatrix} 0.0516 \\ 0.1783 \\ 0.1186 \end{bmatrix}$

EXTENDED KALMAN FILTER

Measurement/correction step

Mean, $\mathrm{X}_1 = \hat{X}_1 + K_1\, v_1 =$

Covariance, $P_1 = \widehat{P_1} - \mathrm{K}_1 \Sigma_{IN_1} K_1^T =$

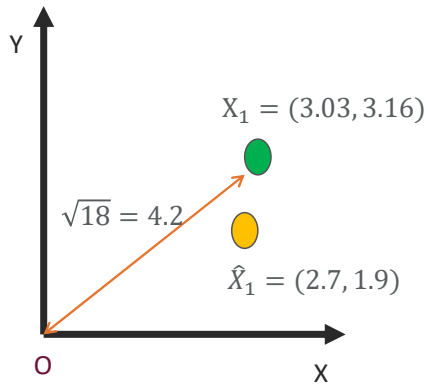


EXTENDED KALMAN FILTER

Measurement/correction step

Mean, $\mathrm{X}_1 = \hat{X}_1 + K_1\, v_1 = \begin{bmatrix} 3.03 \\ 3.16 \\ 1.30 \end{bmatrix}$

Covariance, $P_1 = \widehat{P_1} - \mathrm{K}_1 \Sigma_{IN_1} K_1^T = \begin{bmatrix} 0.56 & -0.77 & -0.87 \\ -0.77 & 1.05 & 1.20 \\ -0.87 & 1.20 & 1.47 \end{bmatrix}$



MARKOV V/S KALMAN FILTER

- Initial belief $bel(x_0)$ also must be a gaussian
  - If the initial robot location is not known: $bel(x_0)$ → Uniform distribution
  - If the initial robot location is perfectly known: $bel(x_0)$ → *Dirac delta* function

Kalman filter allows only gaussian distribution while Markov localization approach allows for arbitrary distribution

- Applications:

  Position tracking: Initial location known → Kalman filter, Markov localization

  Global positioning: Initial location unknown → Markov localization

- Kalman filter is more efficient in high dimensional systems as compared to Markov localization

MAP-BASED LOCALIZATION

Extra reading:

- Chapter 5 of “Introduction to Autonomous Mobile robots” book

  Section 5.6.8 Kalman Filter localization

- Chapter 3 of “Probabilistic Robotics” book

  Section 3.2.2 Kalman Filter